LabGuard+: a Smart Science Laboratory Safety Station with QR Code Equipment Tracking, QR-Based User Log-In Monitoring, and LED Safety Status Display

A Research Proposal
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Background of the Study

Science laboratories serve as essential components of science education, providing students with opportunities to apply theoretical concepts through experimentation, practical activities, and the proper use of laboratory equipment. However, laboratory activities may expose students and laboratory personnel to various hazards involving chemicals, glassware, electrical devices, heat sources, and other potentially dangerous materials. Consequently, maintaining laboratory safety requires not only the availability of appropriate safety equipment but also adequate safety knowledge, proper practices, effective supervision, and systematic monitoring. Hussein (2022) emphasized that laboratory accidents may result from insufficient safety knowledge, unsafe attitudes, and inappropriate laboratory practices among teachers and laboratory personnel. Similarly, Tadesse et al. (2023) reported inconsistencies in the implementation of safety requirements and practices in teaching laboratories, highlighting the need to strengthen laboratory safety management and monitoring mechanisms.

Effective management of laboratory equipment is another essential component of maintaining a safe and organized laboratory environment. Science laboratories contain numerous equipment and materials that require proper identification, storage, monitoring, and accountability. However, the use of manual or paper-based inventory systems may present difficulties in locating equipment, determining its availability, monitoring its condition, and maintaining accurate records, particularly when laboratory facilities are utilized by multiple students and teachers. Mahmudah and Maulidah (2021) noted that the large quantity of laboratory equipment, combined with the limitations of paper-based inventory systems, may create challenges in laboratory management. These limitations indicate the need for more systematic, efficient, and accessible approaches to laboratory equipment identification and tracking.

In addition to equipment management, monitoring laboratory users is important in promoting accountability and responsible use of laboratory facilities. Students and teachers may access laboratory equipment and materials at different times, making accurate documentation of laboratory access necessary. Tjahyaningtijas (2023) examined the application of QR codes for laboratory user attendance and found that the

system provided a more efficient and paperless method of recording laboratory attendance. Similarly, Pantilon (2025) reported that a QR Code identification system improved the accuracy and efficiency of student attendance monitoring in an educational setting. These findings indicate that QR-based identification technologies have potential applications in laboratory environments, particularly for documenting users and strengthening accountability.

The integration of digital technologies into laboratory management may address several limitations associated with conventional monitoring practices. Manual records may make it difficult to determine the current status of equipment, identify users responsible for specific laboratory resources, and maintain accurate documentation during periods of frequent laboratory use. Alvarado (2025) noted that reliance on paper-based records may hinder the efficient monitoring of equipment conditions, user information, and laboratory activities. The adoption of digital monitoring systems may therefore facilitate more organized record-keeping, improve accessibility of information, and reduce the limitations associated with manual documentation.

Despite the development of digital laboratory inventory systems and QR-based monitoring technologies, existing approaches have primarily addressed individual aspects of laboratory management, such as equipment inventory, asset tracking, or user attendance. Meanwhile, visual indicators, including LED-based status displays, have been utilized in various monitoring and safety-related systems to communicate system conditions in an immediate and readily understandable manner (Jadhav, 2024; Ghosh et al., 2022). However, there remains a limited integration of equipment tracking, user access monitoring, and real-time visual safety-status communication within a single laboratory management system. This limitation represents a gap in the development of an integrated and centralized approach to laboratory safety management.

The identified gap is particularly relevant in educational laboratory settings where multiple students and teachers may simultaneously or successively utilize the same facilities and equipment. In the absence of an integrated monitoring mechanism, equipment accountability, user documentation, and safety-status communication may rely on separate manual procedures. Such practices may contribute to incomplete records, difficulty in monitoring equipment usage, and limited awareness of the

laboratory's current status. Furthermore, Hussein and Shifera (2022) emphasized that inadequate safety practices and insufficient training may increase the likelihood of laboratory accidents, reinforcing the importance of systematic safety management and continuous monitoring within laboratory environments.

In response to these identified concerns, the development of an integrated technology-based laboratory safety system may provide a practical approach to improving laboratory monitoring and management. The proposed LabGuard+: Development of a Smart Science Laboratory Safety Station with QR Code Equipment Tracking, QR-Based User Log-In Monitoring, and LED Safety Status Display is designed to integrate three key functions into a single safety station. The QR code equipment tracking feature is intended to facilitate the identification and monitoring of laboratory equipment. The QR-based user log-in monitoring feature is designed to provide an organized record of individuals accessing the laboratory and promote user accountability. Meanwhile, the LED safety status display is intended to provide an immediate and visible indication of the laboratory's current safety status.

Through the integration of these functions, LabGuard+ aims to provide a centralized and technology-based approach to laboratory monitoring and safety management. The proposed system seeks to address existing limitations in equipment tracking, user documentation, and safety-status communication by combining these functions within a single accessible station. Ultimately, the development of LabGuard+ may contribute to a more organized, accountable, and safety-oriented laboratory environment while demonstrating the potential application of digital technologies in improving laboratory management within educational institutions.

Statement of the problem

This study aims to determine the effectiveness of LabGuard+, a smart science laboratory safety station designed to provide QR-based equipment tracking and log-in monitoring, and LED safety status display.

Specifically, it seeks to answer the following questions:

1. How easy and reliable is LabGuard+ to use in terms of:

1.1. Accuracy in logging equipment records and user transactions

1.2. Response time in updating the LED safety status

1.3. Reliability in maintaining complete and consistent database records

1.4. Usability as perceived by users

2. How effective is LabGuard+ in utilizing its integrated features in terms of:

2.1. QR Code-Based Laboratory Equipment Tracking

2.2. QR Code-Based User Log-In Monitoring

2.3. LED Safety Status Display

3. How effective is LabGuard+ in improving the overall efficiency of laboratory safety monitoring, equipment tracking, and user management

4. How can LabGuard+ support laboratory personnel in maintaining laboratory safety, monitoring equipment, and managing laboratory users?

5. What improvements can be made to LabGuard+ to further enhance its effectiveness, reliability, and usability in laboratory safety and monitoring?

Independent Variable (IV): LabGuard+: A Smart Science Laboratory Safety Station with QR Code Equipment Tracking, QR-Based User Log-In Monitoring, and LED Safety Status Display

Dependent Variable (DV): Effectiveness of Laboratory Safety Monitoring, Equipment Tracking, and User Management in terms of ease of use, reliability, accuracy, responsiveness, and usability

Significance of the Study

In this part of the study, the researcher wants to emphasize the benefits of this study to the society

Students - This study may benefit students by promoting responsible and accountable use of laboratory equipment. Through the QR-based equipment tracking system, students may be able to record the borrowing and returning of materials more efficiently. The user log-in monitoring feature may also encourage students to follow proper laboratory entry and exit procedures.

Teachers - The study may assist teachers in monitoring students' laboratory participation and equipment use. The system may provide more organized records of laboratory users and borrowed materials, allowing teachers to identify equipment users and check whether materials have been properly returned. LabGuard+ may also support teachers in reinforcing laboratory safety rules and maintaining a more orderly laboratory environment.

Laboratory Personnel - Laboratory personnels may benefit from a more systematic method of monitoring equipment and users. The QR-based tracking feature may help them determine the availability and movement of laboratory materials, while the digital user records may improve accountability within the laboratory. The LED status display may also help them communicate safety conditions more clearly and efficiently.

Science Laboratory of Arellano University- Jose Rizal Campus - The study may contribute to improving the organization, monitoring, and safety management of the Arellano University -Jose Rizal Campus science laboratory. By integrating equipment tracking, user monitoring, and safety-status communication into one station, LabGuard+ may reduce dependence on paper-based records and support more accessible and efficient laboratory documentation. However, the system will function only as a support tool and will not replace existing safety rules, procedures, teachers, or laboratory personnel.

School Administrators - The findings of this study may provide school administrators with information about the possible use of digital technologies in improving laboratory management. The results of the functionality and usability testing may serve as a basis for evaluating whether similar systems could be considered for other laboratories in the

future. The study may also support the school’s efforts to promote accountability, organization, and technology-assisted safety practices.

Future Researchers - This study may serve as a reference for future researchers who plan to develop or improve smart laboratory management systems. It may provide useful information regarding the design, development, functionality testing, and usability testing of a system that integrates QR-based equipment tracking, QR-based user monitoring, and LED safety-status communication. Future studies may expand the system by including features such as chemical hazard detection, emergency alarms, CCTV monitoring, or consumable-material inventory management.

Theoretical Framework

Technology Acceptance Model and Technology Readiness Index

The Technology Acceptance Model (TAM), originally formulated by Davis in 1989, and the Technology Readiness Index (TRI), established by Parasuraman in 2000, explain how users adopt digital innovations based on perceived ease of use, usefulness, and personal technological readiness. These classical frameworks establish that a person's willingness to embrace new tools depends heavily on cognitive comfort and operational efficiency. In simple terms, these combined models describe how an individual's psychological readiness and perceived utility drive the transition from manual procedures to automated systems.

According to recent empirical literature, this combined framework has proven to be associated with key digital adoption outcomes in automated institutional systems (Sohn and Kwon, 2021; Kelly et al., 2023). Operational convenience and interface clarity are directly linked to reduced user hesitation, minimized logging errors, and increased system utilization (Davis, 1989; Parasuraman, 2000; Sohn and Kwon, 2021). Furthermore, contemporary investigations demonstrate that real-time visual feedback, such as live status displays, enhances user trust and encourages consistent long-term adoption (Kelly et al., 2023).

The relevance of "Technology Acceptance Model and Technology Readiness Index" can be seen in the implementation of the LabGuard+ station, as laboratory personnel transition from paper-based tracking to automated hardware. This study intends to

evaluate how fast QR code scanning lowers operational effort and streamlines daily logbook management. Showing insights into user adoption, this framework establishes the basis for assessing perceived ease of use, visual trust, and overall operational satisfaction.

General Systems Theory

General Systems Theory (GST), pioneered by Ludwig von Bertalanffy in 1968, posits that complex structures consist of interconnected components functioning as an integrated whole through Input-Process-Output (IPO) mechanisms. In simple terms, this concept describes how inputs, processing operations, and outputs function together as a unified system rather than isolated parts. It emphasizes that system efficiency depends on the seamless coordination between physical mechanisms and control logic.

According to system engineering evaluations, this theory has proven to be associated with technical reliability and operational speed in hardware-software architectures (Adams et al., 2022). Input-process-output mechanisms are linked to continuous data synchronization, processing accuracy, and central database management (von Bertalanffy, 1968; Adams et al., 2022). Furthermore, centralizing multiple peripheral modules under a single control unit prevents systemic bottlenecks and hardware failures (Adams et al., 2022).

The relevance of "General Systems Theory" can be seen in the structural architecture of the LabGuard+ station, as multiple hardware and software modules operate simultaneously. This study intends to analyze how QR scans and disinfection buttons act as inputs to trigger microcontroller processing and database verification. Showing insights into system coordination, this theory establishes the foundation for evaluating data transmission speed, processing accuracy, and overall hardware-software reliability.

Quick Response (QR) Technology Integration Theory

Quick Response (QR) Technology Integration Theory, originally established by WaveMotif in 1994 and further conceptualized by Soon in 2008, explains how high-density matrix barcodes enable rapid automated data capture and real-time information retrieval. In simple terms, this concept describes how optical

machine-readable data encoding replaces slow manual data entry with instant electronic identification. It emphasizes that automated data scanning lowers human error rates and optimizes real-time tracking in inventory and monitoring systems.

According to recent empirical literature, this technical framework has proven to be associated with increased record accuracy and operational efficiency in automated monitoring stations (Tiwari, 2016; Alhassan et al., 2020). Optical scanning mechanisms are directly linked to faster verification times, streamlined data logging, and reduced administrative delay (Soon, 2008; Tiwari, 2016). Furthermore, contemporary evaluations confirm that two-dimensional barcode integration enhances data security and prevents inventory mismatches in institutional setups (Alhassan et al., 2020).

The relevance of "Quick Response (QR) Technology Integration Theory" can be seen in the core monitoring functions of the LabGuard+ station, as user log-ins and equipment movements are tracked electronically. This study intends to examine how QR code equipment tracking and QR-based user log-in monitoring optimize administrative workflow and maintain precise laboratory records. Showing insights into automated data capture, this theory establishes the foundation for evaluating scanning responsiveness, logging accuracy, and tracking efficiency.

Conceptual Framework

To analyze the study, this research conceptualized the Input-Process-Output (IPO) model to guide the development of LabGuard+: a Smart Science Laboratory Safety Station with QR Equipment Tracking, QR-Based Log-In Monitoring, and LED Safety Status Display.

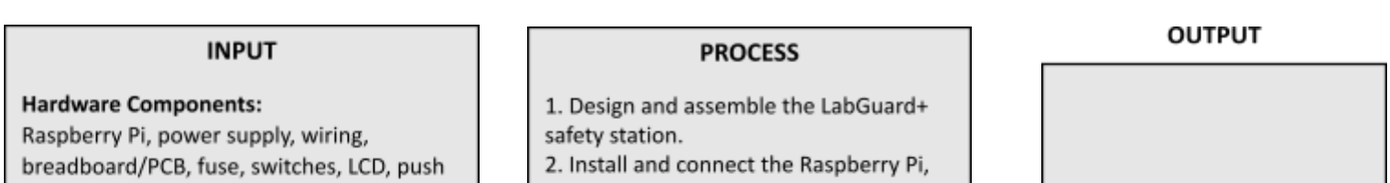




Figure 1 shows the relationship between the Input-Process-Output (IPO) system of the research study. The input includes the hardware, software, programming tools, and system requirements needed to develop the smart science laboratory safety station. These include the Raspberry Pi, QR code scanners, LCD display, LED indicators, UV light, buzzer, power supply, storage compartments, programming software and database. The process involves the design and assembly of the smart science laboratory safety station, programming of the safety status indicators and QR tracking and log-in system, integrating the hardware and software components, and testing the developed system. Through these processes, the study results in the development of LabGuard+: a Smart Science Laboratory Safety Station with QR Code Equipment Tracking, QR-Based User Log-In Monitoring, and LED Safety Status Display.

Scope & Delimitation

This study focuses on the development and evaluation of LabGuard+, a smart safety station designed to improve organization and safety in the science laboratory of Arellano University – Jose Rizal Campus (AU-JRC). The system integrates three main features: (1) QR-based equipment tracking to record the borrowing and returning of laboratory materials, (2) QR-based user log-in monitoring to record the entry and exit of students, teachers, and laboratory personnel, and (3) an LED safety status display to show laboratory safety reminders and status based on system input. The study covers the design, development, functionality testing, and usability testing of the LabGuard+ prototype. The respondents of the study are selected students, teachers, and laboratory personnel of AU-JRC.

This study is limited to the science laboratory of AU-JRC only and will be conducted during School Year 2026–2027. The LabGuard+ system will function solely as a support tool and will not replace teachers, laboratory personnel, or existing laboratory safety rules and procedures. The QR-based equipment tracking feature is limited to equipment with registered QR codes and requires proper scanning to ensure accurate recording of borrowing and returning transactions. The QR-based user log-in monitoring feature requires proper user identification and QR scanning to accurately record laboratory entry and exit and does not provide continuous monitoring of users' activities within the laboratory. The LED display will show pre-programmed safety messages and status indicators and does not provide real-time monitoring. Other laboratories within AU-JRC and other campuses are not included in this study. Furthermore, the system is limited to the three specified features only. Other functions such as chemical hazard detection, emergency alarms, CCTV monitoring, and inventory of consumable materials are not covered by this study

Definition of Terms

The following terms are defined conceptually based on recent related literature and operationally based on their specific use in the development and evaluation of LabGuard+.

Laboratory Management Efficiency - Laboratory management efficiency refers to the organized management and monitoring of laboratory resources, equipment, and records to improve laboratory operations (Adekunle et al., 2024).
In this study, it refers to how LabGuard+ helps manage laboratory equipment, user records, and equipment status.

Laboratory Safety - Laboratory safety refers to practices used to identify and control risks during laboratory activities (Suhanda et al., 2024). In this study, it refers to the safety functions of LabGuard+, particularly the LED safety status display and safety alerts.

LED Safety Status Display - An LED indicator refers to a visual signal used to show the condition or status of a system or equipment. In this study, it refers to the LED indicators of LabGuard+, where green represents SAFE, yellow represents CAUTION, and red represents UNSAFE.

LabGuard+ - A laboratory management system refers to a system used to organize, record, and monitor laboratory resources, equipment, and activities (Cortez et al., 2024). In this study, LabGuard+ refers to the smart science laboratory safety station developed by the researchers. It integrates QR-based equipment tracking, QR-based user log-in monitoring, and LED safety status display.

QR-Based - QR-based technology refers to the use of Quick Response codes to identify and track laboratory equipment and other resources through scanning (Kuswandari et al., 2023). In this study, QR-based refers to the use of QR codes in LabGuard+ to identify users and laboratory equipment, record user log-ins and log-outs, and monitor the borrowing and returning of laboratory materials.

Smart Science Laboratory Safety Station - A smart laboratory system refers to a technology-based system used to improve the management, monitoring, and recording of laboratory equipment and activities (Suhanda et al., 2024). In this study, Smart Science Laboratory Safety Station refers to the LabGuard+ prototype developed to

support laboratory safety and management through QR-based equipment tracking, QR-based user log-in monitoring, and LED safety status display.